

The ₱500 Noche Buena Challenge: A 3×3 Input Pad That Saves More Pins Than Money

If a whole Noche Buena must fit into ₱500, then nine buttons must fit into just a few I/O pins.

Project Objective

The objective of this final project is to design, draft, and present a fully functional hardware module featuring a **3×3 multipurpose button input array** with minimized Arduino I/O usage, a **4-LED BCD indicator output**, and a complete set of design documentation. Students must integrate an appropriate **Combinational Logic Circuit (CLC) IC** to encode inputs and generate BCD signals. In addition, students have the **creative freedom to beautify the input pad**, and the overall aesthetic and usability of the final design will contribute to their grade.

During the project presentation and Arduino testing, the **3×3 pad will function as a 1–9 numeric keypad**, and the **decimal numbers corresponding to the pressed keys will be displayed in the Arduino Serial Monitor**.

Project Description

The module must:

- Accept input from a **3×3 keypad (nine buttons)**.
- Use logic circuitry to reduce required Arduino input pins.
- Display the numeric value of the pressed button using **four LEDs in BCD format (0001–1001)**.
- Send the **decimal number (1–9) to the Arduino Serial Monitor** when a button is pressed.
- Only allow **one button input at a time** to ensure clear and unambiguous readings.

When a button numbered 1 through 9 is pressed, the LEDs must display the corresponding **binary-coded decimal**, and the Arduino Serial Monitor must print the decimal number:

Button	Decimal	BCD (D3 D2 D1 D0)	LED Output
1	1	0001	0001
2	2	0010	0010
3	3	0011	0011

...	...	...	...
9	9	1001	1001

You guys are responsible for selecting the appropriate logic IC(s), such as encoders, multiplexers, or priority encoders, and justifying your selection. The design must clearly show how the keypad feeds the logic system, how the BCD outputs drive the LEDs, and how the reduced number of signals is communicated to the Arduino for serial output.

Key Requirements

1. 3×3 Input Button Array

- Proper electrical configuration and matrix/individual wiring.
- Debouncing strategy and pull-up/down resistor implementation.
- System must register **only one key press at a time**.

2. Logic Circuit Using Chosen CLC IC

- Students must choose any suitable combinational logic IC or IC combination.
- Logic must:
 - Encode the pressed button for Arduino input.
 - Generate the correct 4-bit BCD output for LED indicators.
- Provide truth tables, logic diagrams, or flowcharts demonstrating system behavior.

3. 4-LED BCD Indicator Module

- Four LEDs representing BCD bits (D3–D0).
- LED patterns must match the pressed button's BCD representation (1–9).
- Include proper current limiting and physical placement considerations.

4. Arduino I/O and Serial Output

- Demonstrate how logic encoding reduces the number of required input pins.
- Arduino code must read the encoded signals and display the **decimal number (1–9) in the Serial Monitor**.
- Show that only one button is registered at a time.

5. Drafting, Documentation, and Presentation

- Complete schematic diagram using proper drafting conventions.
- Block diagram showing keypad → logic → LEDs/Arduino/Serial output.
- Full labeling of signals, pinouts, IC orientation, and components.
- PCB layout concept or enclosure sketch.

- Design explanation and testing procedure.
- Project presentation must **demonstrate the keypad operating as a 1–9 numpad**, showing both **LED BCD output and serial number display**.

6. Aesthetic Design & Customization

- Students are encouraged to **beautify and personalize their input pads**.
 - Examples: creative labeling, custom enclosure, color-coded buttons, clean layout, user-friendly arrangement, etc.
- Aesthetic quality, craftsmanship, and overall presentation **will affect the final grade**, rewarding thoughtful and attractive design choices.

7. Testing & Verification Plan

- Students must outline a plan to test:
 - Single keypress detection
 - Proper BCD LED output
 - Arduino Serial Monitor output (decimal 1–9)
 - Include pseudo-code or actual Arduino code if required.
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Deliverables

- Complete drawing set (schematics, diagrams, layout concepts).
- Written explanation covering IC selection and design logic.
- Key-to-BCD mapping and system behavior explanation.
- Final project report and testing plan.
- Final presentation or demonstration showing:
 - **LED BCD outputs**
 - **Serial Monitor decimal output (1–9)**
 - **Beautified input pad design**